18. Let f be continuous on $[a, b]$, let $f(x) \geq 0$ for $x \in [a, b]$, and let $M_n := (\int_a^b f^n)^{1/n}$. Show that $\lim(M_n) = \sup\{f(x) : x \in [a, b]\}$.
19. Suppose that $a > 0$ and that $f \in \mathcal{R}[-a, a]$.
    (a) If f is *even* (that is, if $f(-x) = f(x)$ for all $x \in [0, a]$), show that $\int_{-a}^{a} f = 2\int_0^a f$.
    (b) If f is *odd* (that is, if $f(-x) = -f(x)$ for all $x \in [0, a]$), show that $\int_{-a}^{a} f = 0$.
20. If f is continuous on $[-a, a]$, show that $\int_{-a}^{a} f(x^2)dx = 2\int_0^a f(x^2)dx$.

Section 7.3 The Fundamental Theorem

We will now explore the connection between the notions of the derivative and the integral. In fact, there are *two* theorems relating to this problem: one has to do with integrating a derivative, and the other with differentiating an integral. These theorems, taken together, are called the Fundamental Theorem of Calculus. Roughly stated, they imply that the operations of differentiation and integration are inverse to each other. However, there are some subtleties that should not be overlooked.

The Fundamental Theorem (First Form)

The First Form of the Fundamental Theorem provides a theoretical basis for the method of calculating an integral that the reader learned in calculus. It asserts that if a function f is the derivative of a function F, *and if* f *belongs to* $\mathcal{R}[a, b]$, then the integral $\int_a^b f$ can be calculated by means of the evaluation $F|_a^b := F(b) - F(a)$. A function F such that $F'(x) = f(x)$ for all $x \in [a, b]$ is called an **antiderivative** or a **primitive of** f on $[a, b]$. Thus, when f has an antiderivative, it is a very simple matter to calculate its integral.

In practice, it is convenient to allow some exceptional points c where $F'(c)$ does not exist in $\mathbb{R}$, or where it does not equal $f(c)$. It turns out that we can permit a *finite* number of such exceptional points.

7.3.1 Fundamental Theorem of Calculus (First Form) *Suppose there is a* ***finite*** *set* $\boldsymbol{E}$ *in* $[a, b]$ *and functions* f, $F := [a, b] \to \mathbb{R}$ *such that*:

(a) F *is continuous on* $[a, b]$,

(b) $F'(x) = f(x)$ *for all* $x \in [a, b] \backslash E$,

(c) f *belongs to* $\mathcal{R}[a, b]$.

Then we have

$$\int_a^b f = F(b) - F(a). \tag{1}$$

Proof. We will prove the theorem in the case where $E := \{a, b\}$. The general case can be obtained by breaking the interval into the union of a finite number of intervals (see Exercise 1).

Let $\varepsilon > 0$ be given. Since $f \in \mathcal{R}[a, b]$ by assumption (c), there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is *any* tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then

$$\left| S(f; \dot{\mathcal{P}}) - \int_a^b f \right| < \varepsilon. \tag{2}$$

If the subintervals in $\dot{\mathcal{P}}$ are $[x_{i-1}, x_i]$, then the Mean Value Theorem 6.2.4 applied to F on $[x_{i-1}, x_i]$ implies that there exists $u_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = F'(u_i) \cdot (x_i - x_{i-1}) \qquad \text{for} \qquad i = 1, \ldots, n.$$

If we add these terms, note the telescoping of the sum, and use the fact that $F'(u_i) = f(u_i)$, we obtain

$$F(b) - F(a) = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) = \sum_{i=1}^{n} f(u_i)(x_i - x_{i-1}).$$

Now let $\dot{\mathcal{P}}_u := \{([x_{i-1}, x_i], u_i)\}_{i=1}^{n}$, so the sum on the right equals $S(f; \dot{\mathcal{P}}_u)$. If we substitute $F(b) - F(a) = S(f; \dot{\mathcal{P}}_u)$ into (2), we conclude that

$$\left| F(b) - F(a) - \int_a^b f \right| < \varepsilon.$$

But, since $\varepsilon > 0$ is arbitrary, we infer that equation (1) holds. Q.E.D.

Remark If the function F is differentiable at every point of $[a, b]$, then (by Theorem 6.1.2) hypothesis (a) is automatically satisfied. If f is not defined for some point $c \in E$, we take $f(c) := 0$. Even if F is differentiable at every point of $[a, b]$, condition (c) is *not automatically satisfied*, since there exist functions F such that F' is not Riemann integrable. (See Example 7.3.2(e).)

7.3.2 Examples **(a)** If $F(x) := \frac{1}{2}x^2$ for all $x \in [a, b]$, then $F'(x) = x$ for all $x \in [a, b]$. Further, $f = F'$ is continuous so it is in $\mathcal{R}[a, b]$. Therefore the Fundamental Theorem (with $E = \emptyset$) implies that

$$\int_a^b x\, dx = F(b) - F(a) = \tfrac{1}{2}(b^2 - a^2).$$

(b) If $G(x) := \operatorname{Arctan} x$ for $x \in [a, b]$, then $G'(x) = 1/(x^2 + 1)$ for all $x \in [a, b]$; also G' is continuous, so it is in $\mathcal{R}[a, b]$. Therefore the Fundamental Theorem (with $E = \emptyset$) implies that

$$\int_a^b \frac{1}{x^2 + 1}\, dx = \operatorname{Arctan} b - \operatorname{Arctan} a.$$

(c) If $A(x) := |x|$ for $x \in [-10, 10]$, then $A'(x) = -1$ if $x \in [-10, 0)$ and $A'(x) = +1$ for $x \in (0, 10]$. Recalling the definition of the signum function (in 4.1.10(b)), we have $A'(x) = \operatorname{sgn}(x)$ for all $x \in [-10, 10] \backslash \{0\}$. Since the signum function is a step function, it belongs to $\mathcal{R}[-10, 10]$. Therefore the Fundamental Theorem (with $E = \{0\}$) implies that

$$\int_{-10}^{10} \operatorname{sgn}(x)\, dx = A(10) - A(-10) = 10 - 10 = 0.$$

(d) If $H(x) := 2\sqrt{x}$ for $x \in [0, b]$, then H is continuous on $[0, b]$ and $H'(x) = 1/\sqrt{x}$ for $x \in (0, b]$. Since $h := H'$ is not bounded on $(0, b]$, it does not belong to $\mathcal{R}[0, b]$ no matter how we define $h(0)$. Therefore, the Fundamental Theorem 7.3.1 does not apply. (However, we will see in Example 10.1.10(a) that h is *generalized* Riemann integrable on $[0, b]$.)

(e) Let $K(x) := x^2\cos(1/x^2)$ for $x \in (0, 1]$ and let $K(0) := 0$. It follows from the Product Rule 6.1.3(c) and the Chain Rule 6.1.6 that

$$K'(x) = 2x\cos(1/x^2) + (2/x)\sin(1/x^2) \qquad \text{for} \qquad x \in (0, 1].$$

Further, as in Example 6.1.7(e), it can be shown that $K'(0) = 0$. Thus K is continuous and differentiable at *every point* of $[0, 1]$. Since it can be seen that the function K' is not bounded on $[0, 1]$, it does not belong to $\mathcal{R}[0, 1]$ and the Fundamental Theorem 7.3.1 does not apply to K'. (However, we will see from Theorem 10.1.9 that K' is *generalized* Riemann integrable on $[0, 1]$.) □

The Fundamental Theorem (Second Form)

We now turn to the Fundamental Theorem (Second Form) in which we wish to differentiate an integral involving a variable upper limit.

7.3.3 Definition If $f \in \mathcal{R}[a, b]$, then the function defined by

$$F(z) := \int_a^z f \qquad \text{for} \qquad z \in [a, b], \tag{3}$$

is called the **indefinite integral** of f with **basepoint** a. (Sometimes a point other than a is used as a basepoint; see Exercise 6.)

We will first show that if $f \in \mathcal{R}[a, b]$, then its indefinite integral F satisfies a Lipschitz condition; hence F is continuous on $[a, b]$.

7.3.4 Theorem *The indefinite integral F defined by* (3) *is continuous on* $[a, b]$. *In fact, if* $|f(x)| \leq M$ *for all* $x \in [a, b]$, *then* $|F(z) - F(w)| \leq M|z - w|$ *for all* $z, w \in [a, b]$.

Proof. The Additivity Theorem 7.2.9 implies that if $z, w \in [a, b]$ and $w \leq z$, then

$$F(z) = \int_a^z f = \int_a^w f + \int_w^z f = F(w) + \int_w^z f,$$

whence we have

$$F(z) - F(w) = \int_w^z f.$$

Now if $-M \leq f(x) \leq M$ for all $x \in [a, b]$, then Theorem 7.1.5(c) implies that

$$-M(z - w) \leq \int_w^z f \leq M(z - w),$$

whence it follows that

$$|F(z) - F(w)| \leq \left|\int_w^z f\right| \leq M|z - w|,$$

as asserted. Q.E.D.

We will now show that the indefinite integral F is differentiable at any point where f is continuous.

7.3.5 Fundamental Theorem of Calculus (Second Form) *Let $f \in \mathcal{R}[a,b]$ and let f be continuous at a point $c \in [a,b]$. Then the indefinite integral, defined by* (3)*, is differentiable at c and $F'(c) = f(c)$.*

Proof. We will suppose that $c \in [a,b)$ and consider the right-hand derivative of F at c. Since f is continuous at c, given $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $c \leq x < c + \eta_\varepsilon$, then

$$f(c) - \varepsilon < f(x) < f(c) + \varepsilon. \tag{4}$$

Let h satisfy $0 < h < \eta_\varepsilon$. The Additivity Theorem 7.2.9 implies that f is integrable on the intervals $[a,c]$, $[a,\ c+h]$ and $[c,\ c+h]$ and that

$$F(c+h) - F(c) = \int_c^{c+h} f.$$

Now on the interval $[c,\ c+h]$ the function f satisfies inequality (4), so that we have

$$(f(c) - \varepsilon) \cdot h \leq F(c+h) - F(c) = \int_c^{c+h} f \leq (f(c) + \varepsilon) \cdot h.$$

If we divide by $h > 0$ and subtract $f(c)$, we obtain

$$\left| \frac{F(c+h) - F(c)}{h} - f(c) \right| \leq \varepsilon.$$

But, since $\varepsilon > 0$ is arbitrary, we conclude that the right-hand limit is given by

$$\lim_{x \to 0+} \frac{F(c+h) - F(c)}{h} = f(c).$$

It is proved in the same way that the left-hand limit of this difference quotient also equals $f(c)$ when $c \in (a,\ b]$, whence the assertion follows. Q.E.D.

If f is continuous on all of $[a,\ b]$, we obtain the following result.

7.3.6 Theorem *If f is continuous on $[a,\ b]$, then the indefinite integral F, defined by* (3)*, is differentiable on $[a,\ b]$ and $F'(x) = f(x)$* for all $x \in [a,b]$.

Theorem 7.3.6 can be summarized: *If f is continuous on $[a,\ b]$, then its indefinite integral is an antiderivative of f.* We will now see that, in general, the indefinite integral need not be an antiderivative (either because the derivative of the indefinite integral does not exist or does not equal $f(x)$).

7.3.7 Examples **(a)** If $f(x) := \operatorname{sgn} x$ on $[-1,\ 1]$, then $f \in \mathcal{R}[-1,\ 1]$ and has the indefinite integral $F(x) := |x| - 1$ with the basepoint -1. However, since $F'(0)$ does not exist, F is not an antiderivative of f on $[-1,\ 1]$.

(b) If h denotes Thomae's function, considered in 7.1.7, then its indefinite integral $H(x) := \int_0^x h$ is identically 0 on [0, 1]. Here, the derivative of this indefinite integral exists at every point and $H'(x) = 0$. But $H'(x) \neq h(x)$ whenever $x \in \mathbb{Q} \cap [0,\ 1]$, so that H is not an antiderivative of h on [0, 1]. □